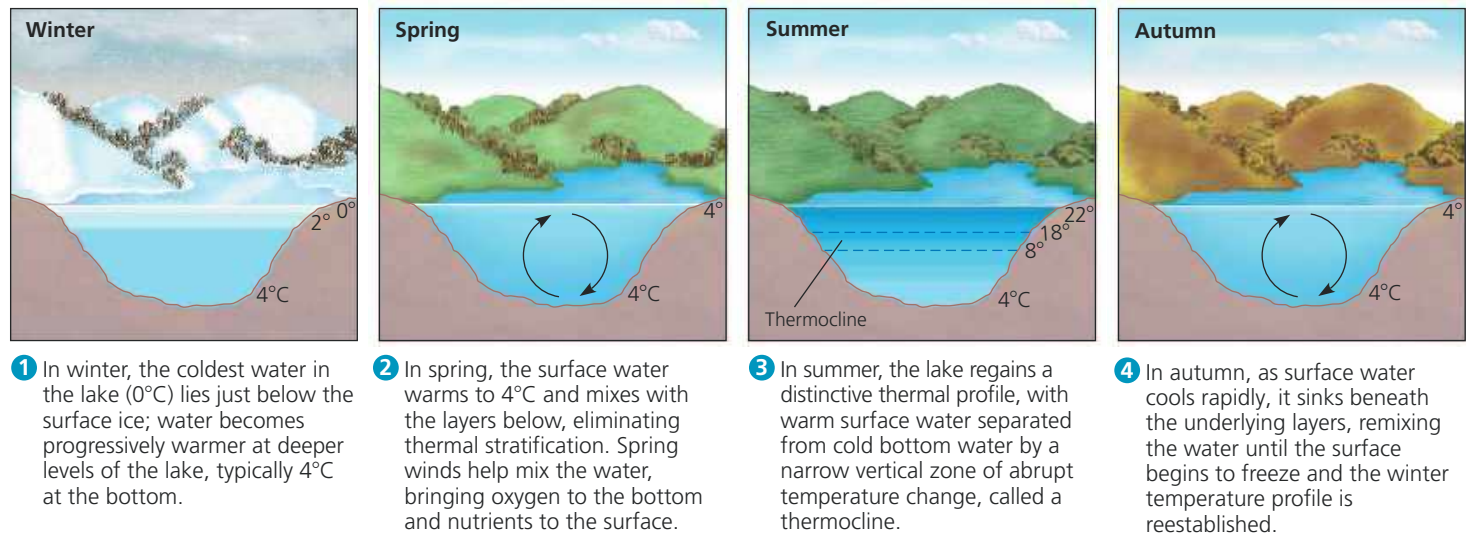


▼ **Figure 52.15 Seasonal turnover in lakes with winter ice cover.** Seasonal turnover causes the lake waters to be well oxygenated at all depths in spring and autumn; in winter and summer, when the lake is stratified by temperature, the oxygen concentration decreases with depth.



semiannual mixing of their waters as a result of changing temperature profiles (Figure 52.15). This **turnover**, as it is called, sends oxygenated water from a lake's surface to the bottom and brings nutrient-rich water from the bottom to the surface in both spring and autumn.

In both freshwater and marine environments, communities are distributed according to water depth, degree of light penetration, distance from shore, and whether they are found in open water or near the bottom. Marine communities, in particular, illustrate the limitations on species distribution that result from these abiotic factors. Plankton and many fish species live in the relatively shallow photic zone (see Figure 52.14b). Because water absorbs light so well and the ocean is so deep, most of the ocean volume is dark (the aphotic zone) and harbors relatively little life.

Figure 52.16 explores the main characteristics of Earth's major aquatic biomes.

CONCEPT CHECK 52.3

1. Why are phytoplankton, and not benthic algae or rooted aquatic plants, the dominant photosynthetic organisms of the oceanic pelagic zone? (See Figure 52.16.)
2. **MAKE CONNECTIONS** Many organisms living in estuaries experience freshwater and saltwater conditions each day with the rising and falling of tides. Explain how these changing conditions challenge the survival of these organisms (see Concept 44.1).
3. **MAKE CONNECTIONS** As noted in Figure 52.16, the addition of nutrients to a lake can cause an algal bloom. When these algae die, complex molecules in their bodies are broken down by decomposers using aerobic respiration. Explain why this could reduce the lake's oxygen levels (see Concept 9.1).

For suggested answers, see Appendix A.

CONCEPT 52.4

Interactions between organisms and the environment limit the distribution of species

Broadly speaking, species distributions are a consequence of both ecological factors and evolutionary history. Consider kangaroos, which are found in Australia and nowhere else in the world. Fossil evidence indicates that kangaroos and their close relatives originated in Australia, roughly 5 million years ago. By that time, Australia had moved close to its present location (by continental drift; see Concept 25.4), and it was not connected to other land-masses. Thus, kangaroos occur only in Australia in part because of an accident of history: The kangaroo lineage originated there at a point in time when the continent was geographically isolated.

But ecological factors are also important. To date, kangaroos have not dispersed (on their own) to other continents; hence, they are restricted to the continent on which they originated. And within Australia, kangaroos are found in some habitats but not in others. The red kangaroo, for example, occurs in the arid grasslands of central Australia, but not in the tall, open forests of eastern Australia. Moreover, kangaroos are not unusual in this respect—all species are found in some habitats but not others. Hence, ecologists ask not only *where* species occur, but also *why* species occur where they do: What ecological factors—biotic and abiotic—determine their distribution?

In many cases, both biotic and abiotic factors affect the distribution of a species, as is true of the saguaro cactus (*Carnegiea gigantea*). Saguaros are found almost exclusively in the Sonoran Desert of the southwestern United States and